

The dual cone of a polyhedral cone

Kent Andersen

In this note we characterize the dual cone of a polyhedral cone. The dual cone of a convex cone describes the supporting inequalities for the cone. To describe a supporting inequality for a convex cone, it suffices to record the normal vector of the inequality. Indeed, every supporting inequality for a convex cone has zero on the right-hand-side (proved in note 1.(B)).

Fact 1 (*All supporting inequalities for convex cones support the origin*)
The inequality $\alpha^T x \leq \beta$ is supporting for a convex cone $K \subseteq \mathbb{R}^n \iff \beta = 0$, and the inequality $\alpha^T x \leq 0$ is valid for K . ■

The set of normal vectors for the supporting inequalities for a convex cone is called the *dual* of the convex cone.

Definition 1 (*Definition of the dual cone of a convex cone*)
For a convex cone $K \subseteq \mathbb{R}^n$, the dual cone K° of K is the set:
$$K^\circ := \{\alpha \in \mathbb{R}^n : \alpha^T x \leq 0 \text{ is valid for } K\}.$$
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We next characterize the dual cone C° of a polyhedral cone $C \subseteq \mathbb{R}^n$. We assume that the polyhedral cone C is given as the solution set for a finite number of linear inequalities with zero on the right-hand-side:

$$C = \{x \in \mathbb{R}^n : Ax \leq 0_m\},$$

where $A \in \mathbb{R}^{m \times n}$, and m is the (finite) number of inequalities. The vectors $a_1, \dots, a_m$ denote the rows of A .

The key in the characterization of C° is to use the fact that finitely generated convex cones are polyhedral cones (proved in note 1.(D)). In particular this implies that $\text{cone}(\{a_1, \dots, a_m\})$ is a polyhedral cone.

Fact 2 (*Finitely generated cones are polyhedral cones*)
One has $\text{cone}(\{a_1, \dots, a_m\}) = \{x \in \mathbb{R}^n : Dx \leq 0_p\}$ for some $D \in \mathbb{R}^{p \times n}$. ■

In the remainder the rows of D are denoted $d_1, \dots, d_p$. The vectors $d_1, \dots, d_p$ have the following very important property.

Observation 1 (*Property of the inequalities describing $\text{cone}(\{a_1, \dots, a_m\})$*)
One has $d_1, \dots, d_p \in C$. ■

Indeed, Fact 2 combined with $a_1, \dots, a_m \in \text{cone}(\{a_1, \dots, a_m\})$ shows $Da_i \leq 0_p$ holds for all $i \in \{1, \dots, m\}$, or equivalently one has $DA^T \leq 0_{p \times m}$. Transposing both sides of this inequality gives $AD^T \leq 0_{m \times p}$, or equivalently $Ad_k \leq 0_m$ for all $k \in \{1, \dots, p\}$, which shows $d_1, \dots, d_p \in C$.

Observation 1 gives the following characterization of C° . Observe that non-negative combinations of the inequalities $Ax \leq 0_m$ give inequalities that support C , *i.e.*, one has $\text{cone}(\{a_1, \dots, a_m\}) \subseteq C^\circ$. The fact is that *all* supporting inequalities have this form, *i.e.*, one has $\text{cone}(\{a_1, \dots, a_m\}) = C^\circ$.

Theorem 1 (*Characterization of the dual cone of a polyhedral cone*)
The dual cone C° of C is given by:

$$C^\circ = \text{cone}(\{a_1, \dots, a_m\}). \quad \blacksquare$$

Proof: The inclusion $\text{cone}(\{a_1, \dots, a_m\}) \subseteq C^\circ$ holds since non-negative combinations of the inequalities $Ax \leq 0_m$ give inequalities that support C .

To see $C^\circ \subseteq \text{cone}(\{a_1, \dots, a_m\})$, let $\delta \in C^\circ$ be arbitrary. We will show $\delta \in \text{cone}(\{a_1, \dots, a_m\})$. Observation 1 gives $d_1, \dots, d_p \in C$. By definition of C° this implies $\delta^T d_i \leq 0$ for all $i \in \{1, \dots, p\}$, or equivalently $D\delta \leq 0_p$. Hence $\delta \in \{x \in \mathbb{R}^n : Dx \leq 0_p\} = \text{cone}(\{a_1, \dots, a_m\})$. $\blacksquare$

The dual of a convex cone is again a convex cone: Any conic combination of normal vectors that define supporting inequalities for a convex cone $K \subseteq \mathbb{R}^n$ is again a vector that defines a supporting inequality for K . A dual cone $(K^\circ)^\circ$ of the dual cone K° of K therefore exists.

A key property of a polyhedral cone is that the dual of the dual cone is the original cone.

Theorem 2 (*The dual of the dual of a polyhedral cone is the original cone*)
The polyhedral cone C satisfies $(C^\circ)^\circ = C$. $\blacksquare$

Proof: Theorem 1 shows $C^\circ = \text{cone}(\{a_1, \dots, a_m\})$.

$\subseteq$: First suppose $\bar{x} \in (C^\circ)^\circ$. Since $C^\circ = \text{cone}(\{a_1, \dots, a_m\})$ and $(C^\circ)^\circ$ is the set of supporting inequalities for C° , this implies $a_i^T \bar{x} \leq 0$ for all $i \in \{1, \dots, m\}$, and therefore $\bar{x} \in C$.

$\supseteq$: Now suppose $\gamma \in C$. This implies $\gamma^T a_i \leq 0$ for all $i \in \{1, \dots, m\}$, and therefore also $\gamma^T x \leq 0$ for all $x \in \text{cone}(\{a_1, \dots, a_m\})$. Since $C^\circ = \text{cone}(\{a_1, \dots, a_m\})$, this implies $\gamma^T x \leq 0$ supports C° . Hence $\gamma \in (C^\circ)^\circ$. $\blacksquare$